



IOTRICITY S3

Team Name

ChargeX

Problem Statement

Homes, farms, industries rely on manual tank checks, leading to overflow, water wastage, and pump damage from dry runs due to no real-time visibility



Problem Identification & Analysis

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What is the real-world problem?

- Water tanks overflow or run dry unnoticed
- Manual checking wastes time and water
- Motor left ON/OFF by guesswork burns pumps



Who faces it, and where?

- Households, apartments and hostels
- Farmers and irrigation tanks
- Schools, factories and small industries



How is it solved today?

- Visual checks and manual switching
- Float switches / basic mechanical valves
- Costly commercial automation systems



Gaps in existing solutions

- No remote view of the water level
- No low-water or overflow alerts
- Expensive, hard to install or customize

Identified Problem

“Water tanks are monitored and refilled manually, causing overflow, dry-running pumps and wasted water, with no remote visibility or alerts.”

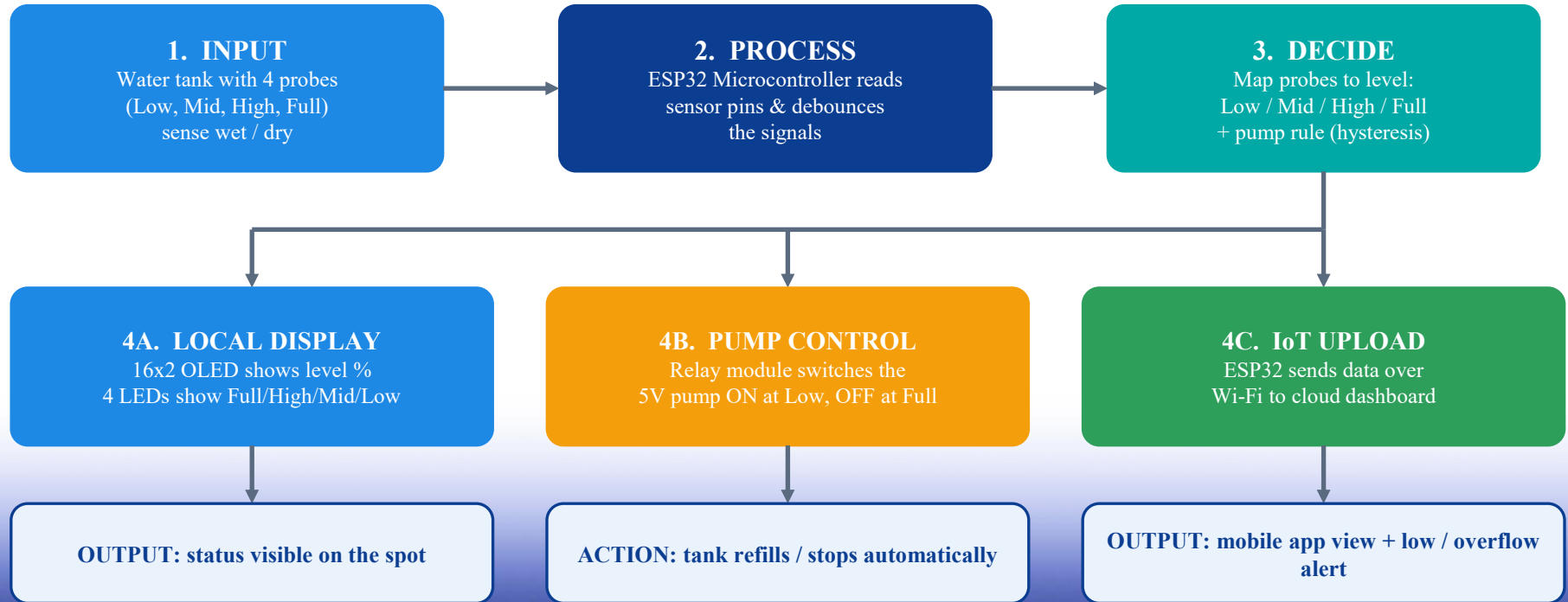
Our Proposed Solution

“We propose to develop an IoT-based water level monitoring and automatic pump control system that will prevent overflow and dry-run, save water, and send real-time alerts to a mobile app.”

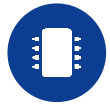


Proposed Technical Approach — Working Flow

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Feedback loop: rising water level is sensed again by the probes — the cycle repeats continuously



Technology Stack

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HARDWARE



Water Level Ultrasonic Sensors

4 probes (Top, High, Mid, Low) detect water at each level



ESP32 Microcontroller & Buzzer (audio alert)

Microcontroller: reads sensors, runs logic, drives outputs; Buzzer triggers sound alert for full tank detection.



ESP32 Wi-Fi

Sends level data to cloud and mobile app



Relay Module

Safely switches the pump ON/OFF from low-voltage signals



Water Pump (5V)

Fills the tank automatically



16x2 OLED

Shows live level and status locally



LED Indicators

Red / Yellow / Green / Blue tank status



5V DC Supply

Stable power for the circuit and pump



Tank, Board, Wires

Transparent tank, base board, jumpers, tubing

SOFTWARE



Arduino IDE + C/C++

Write and upload the control code



Libraries

LiquidCrystal_I2C, ESP32WiFi, IoT platform library



IoT Cloud Platform

e.g. Blynk / ThingSpeak stores and streams data



Mobile App / Dashboard

Live level, status, history graph, alerts



Feasibility & Viability

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Technical Feasibility

- Uses common, proven modules
- Simple digital I/O + Wi-Fi upload
- Buildable with basic Arduino skills



Resources Required

- Components listed in Tech Stack
- Laptop, ESP32 Microcontroller , Wi-Fi hotspot
- Multimeter, soldering / breadboard



Budget (approx.)

- ESP32 Microcontroller ~ ₹300
- Pump, relay, LCD, probes ~ ₹720
- Tank, supply, wires, LEDs ~ ₹500
- Buzzer/Alarm Module ~ ₹50
- Total: about ₹1500 - 2,000



Time

- Wk 1: design & procure parts
- Wk 2: wiring & coding
- Wk 3: IoT app & testing
- Fits a 3–4 week timeline



Challenges → Fixes

- Probe corrosion → use SS probes
- Wi-Fi drops → local LCD/OLED backup
- Pump dry-run → low-level cutoff



Future Scope

- Easy to maintain: low-cost parts
- Modular: add sensors or tanks
- Scales to homes, farms, industry



Impact & Benefits

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Homes & Apartments



Farmers



Industry & Factories



Schools & Communities



Key Benefits

- Time: no manual tank checks
- Cost: avoids wasted water & pump wear
- Safety: prevents overflow damage
- Convenience: monitor from anywhere



Expected Impact

- Less water wastage
- Longer pump life
- Smarter, data-driven water use
- Low-cost model others can copy



Impact Measurement

- Overflow / dry-run events before vs after
- Water & electricity use per week
- Alert response time in app
- User feedback from test households



Limitations & Future

- Limits: needs Wi-Fi; probes need cleaning
- Add ultrasonic/flow sensors, solar power
- Add SMS alerts and usage prediction



Level Logic: Sensors → Action

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Tank State	Probes Wet	LED	Pump	App / Alert
Empty	0%	No Light	OFF	Critical low alert
Low	0-25%	Red	ON (filling)	Low water alert
Mid	25-50%	Yellow	ON (filling)	Status: Normal
High	50-75%	Green	ON (filling)	Status: Normal
Almost Full to Full	75-100%	Blue	OFF	Tank full / overflow alert

Control Logic (pseudo-code)

```
if level == LOW → pump ON
if level == FULL → pump OFF
update LCD, OLEDs, then upload via ESP32
```

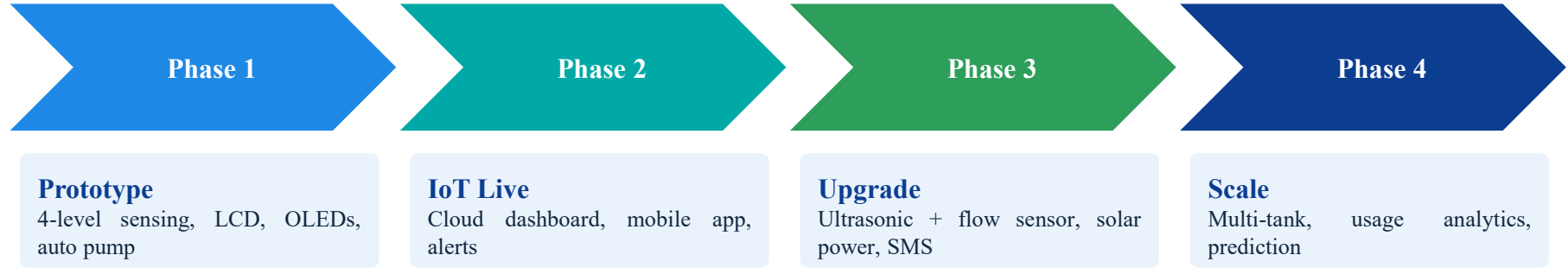
Why hysteresis?

The pump starts at Low and stops only at Full, so it does not rapidly switch ON/OFF when the water ripples. This protects the relay and pump.



Roadmap & Conclusion

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- **Wokwi simulation (Circuit + Code):** <https://wokwi.com/projects/476309690117978113>
- **YouTube:** <https://youtu.be/GHWsvLkek4I?si=gipui1CGjjyzn43D>
- **GitHub Repository:** <https://github.com/Kriti2706/Smart-water-managements.git>

Conclusion

A low-cost, IoT-enabled system that senses the water level, controls the pump automatically and alerts users in real time — saving water, time and money for homes, farms and industry.

Save Water | Smart Monitoring | Better Tomorrow
Thank You!